```
// ⍰ A) Armstrong Number Program
using System;
class Armstrong
{
    static void Main()
    {
        int num, sum = 0, temp, rem;
        Console.Write("Enter a number: ");
        num = Convert.ToInt32(Console.ReadLine());
        temp = num;
        while (num != 0)
        {
            rem = num % 10;
            sum += rem * rem * rem;
            num /= 10;
        }
        Console.WriteLine(temp == sum ? "Armstrong Number" : "Not
an Armstrong Number");
    }
}

// ⍰ B) Staff Administration System using Arrays
using System;
class StaffAdmin
```

```
{
    static void Main()
    {
        string[] names = new string[100];
        string[] positions = new string[100];
        int count = 0;
        while (true)
        {
            Console.WriteLine("\n1. Add Staff\n2. Delete Staff\n3. Display
Staff\n4. Exit");
            Console.Write("Enter choice: ");
            int ch = Convert.ToInt32(Console.ReadLine());
            if (ch == 1)
            {
                Console.Write("Enter Name: ");
                names[count] = Console.ReadLine();
                Console.Write("Enter Position: ");
                positions[count] = Console.ReadLine();
                count++;
            }
            else if (ch == 2)
            {
                Console.Write("Enter Name to delete: ");
                string name = Console.ReadLine();
```

```
            for (int i = 0; i < count; i++)
            {
                if (names[i] == name)
                {
                    for (int j = i; j < count - 1; j++)
                    {
                        names[j] = names[j + 1];
                        positions[j] = positions[j + 1];
                    }
                    count--;
                    Console.WriteLine("Deleted successfully");
                    break;
                }
            }
        }
        else if (ch == 3)
        {
            for (int i = 0; i < count; i++)
                Console.WriteLine("Name: {0}, Position: {1}", names[i],
positions[i]);
        }
        else
            break;
    }
```

```
    }
}

// ⍰ A) Leap Year Check
using System;
class LeapYear
{
    static void Main()
    {
        Console.Write("Enter a year: ");
        int year = Convert.ToInt32(Console.ReadLine());
        if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
            Console.WriteLine("Leap Year");
        else
            Console.WriteLine("Not a Leap Year");
    }
}

// ⍰ B) Remove Spaces with Hyphens using Delegates
using System;
delegate string StringModifier(string input);
class ReplaceSpaces
{
```

```
    static string Replace(string str)
    {
        return str.Replace(' ', '-');
    }

    static void Main()
    {
        Console.Write("Enter a string: ");
        string input = Console.ReadLine();
        StringModifier del = new StringModifier(Replace);
        string result = del(input);
        Console.WriteLine("Modified string: " + result);
    }
}

// □ A) Fibonacci Series
using System;
class Fibonacci
{
    static void Main()
    {
        int n1 = 0, n2 = 1, n3, i, num;
        Console.Write("Enter number of terms: ");
```

```
        num = Convert.ToInt32(Console.ReadLine());
        Console.Write(n1 + " " + n2 + " ");
        for (i = 2; i < num; ++i)
        {
            n3 = n1 + n2;
            Console.Write(n3 + " ");
            n1 = n2;
            n2 = n3;
        }
    }
}

// ⍰ B) Get five marks and calculate total, percentage, average
using System;
class StudentMarks
{
    static void Main()
    {
        int[] marks = new int[5];
        int total = 0;
        for (int i = 0; i < 5; i++)
        {
            Console.Write("Enter mark {0}: ", i + 1);
```

```
            marks[i] = Convert.ToInt32(Console.ReadLine());
            total += marks[i];
        }
        float average = total / 5.0f;
        float percentage = (total / 500.0f) * 100;
        Console.WriteLine("Total: " + total);
        Console.WriteLine("Average: " + average);
        Console.WriteLine("Percentage: " + percentage);
    }
}

// ⍰ A) Even numbers from 1 to 100
using System;
class EvenNumbers
{
    static void Main()
    {
        for (int i = 1; i < 100; i++)
        {
            if (i % 2 == 0)
                Console.Write(i + " ");
        }
    }
```

```
}

// B) Area of Square, Rectangle and Triangle using Method Overloading
using System;
class ShapeArea
{
    static double Area(double side)
    {
        return side * side;
    }

    static double Area(double length, double width)
    {
        return length * width;
    }

    static double Area(double b, double h, bool isTriangle)
    {
        return 0.5 * b * h;
    }

    static void Main()
    {
```

```
        Console.WriteLine("Area of Square (5): " + Area(5));
        Console.WriteLine("Area of Rectangle (4,6): " + Area(4, 6));
        Console.WriteLine("Area of Triangle (base 4, height 5): " +
Area(4, 5, true));
    }
}

// ⍰ A) Sort names in descending order
using System;
using System.Linq;
class NameSorter
{
    static void Main()
    {
        string[] names = { "Zara", "Maya", "Asha", "John" };
        var sorted = names.OrderByDescending(n => n);
        foreach (string name in sorted)
            Console.WriteLine(name);
    }
}

// ⍰ B) LINQ - Add, Filter and Display Students with Grade > Value
using System;
using System.Collections.Generic;
```

```
using System.Linq;
class Student
{
    public string Name { get; set; }
    public int Age { get; set; }
    public double Grade { get; set; }
}

class StudentLINQ
{
    static void Main()
    {
        List<Student> students = new List<Student>();
        students.Add(new Student { Name = "John", Age = 20, Grade =
85 });
        students.Add(new Student { Name = "Alice", Age = 22, Grade =
75 });
        students.Add(new Student { Name = "Bob", Age = 19, Grade = 90
});

        Console.Write("Enter minimum grade to filter: ");
        double minGrade = Convert.ToDouble(Console.ReadLine());

        var filtered = students.Where(s => s.Grade > minGrade);
```

```
        foreach (var s in filtered)
        {
            Console.WriteLine("Name: {0}, Age: {1}, Grade: {2}", s.Name,
s.Age, s.Grade);
        }
    }
}

// □ A) Palindrome Check
using System;
class PalindromeCheck
{
    static void Main()
    {
        Console.Write("Enter a string: ");
        string input = Console.ReadLine();
        string reverse = new string(input.Reverse().ToArray());
        Console.WriteLine(input == reverse ? "Palindrome" : "Not a
Palindrome");
    }
}

// □ B) Lambda: Sort and Filter Employee List
using System;
```

```
using System.Collections.Generic;
using System.Linq;
class Employee
{
    public string Name { get; set; }
    public int Age { get; set; }
    public int YearsOfExperience { get; set; }
}

class LambdaEmployee
{
    static void Main()
    {
        List<Employee> employees = new List<Employee>()
        {
            new Employee { Name = "John", Age = 30, YearsOfExperience
= 6 },
            new Employee { Name = "Asha", Age = 24, YearsOfExperience
= 3 },
            new Employee { Name = "Raj", Age = 29, YearsOfExperience =
7 }
        };

        var sorted = employees.OrderBy(e => e.Age);
```

```
        Console.WriteLine("Sorted by Age:");
        foreach (var e in sorted)
            Console.WriteLine("{0}, Age: {1}", e.Name, e.Age);

        var filtered = employees.Where(e => e.YearsOfExperience > 5);
        Console.WriteLine("\nMore than 5 years experience:");
        foreach (var e in filtered)
            Console.WriteLine(e.Name);
    }
}

// □ A) Swap Two Numbers Without Using Third Variable
using System;
class SwapNumbers
{
    static void Main()
    {
        Console.Write("Enter first number: ");
        int a = Convert.ToInt32(Console.ReadLine());
        Console.Write("Enter second number: ");
        int b = Convert.ToInt32(Console.ReadLine());

        a = a + b;
```

```
        b = a - b;
        a = a - b;

        Console.WriteLine("After swapping: a = {0}, b = {1}", a, b);
    }
}

// □ B) Calculate perimeter and area using loops
using System;
class ShapeCalculator
{
    static void Main()
    {
        while (true)
        {
            Console.WriteLine("\n1. Circle\n2. Triangle\n3. Square\n4.
Exit");
            Console.Write("Enter your choice: ");
            int choice = Convert.ToInt32(Console.ReadLine());

            if (choice == 1)
            {
                Console.Write("Enter radius: ");
                double r = Convert.ToDouble(Console.ReadLine());
```

```
            Console.WriteLine("Area: {0}", Math.PI * r * r);
            Console.WriteLine("Perimeter: {0}", 2 * Math.PI * r);
        }
        else if (choice == 2)
        {
            Console.Write("Enter base and height: ");
            double b = Convert.ToDouble(Console.ReadLine());
            double h = Convert.ToDouble(Console.ReadLine());
            Console.WriteLine("Area: {0}", 0.5 * b * h);
            Console.WriteLine("Perimeter (approx): {0}", b + h +
Math.Sqrt(b * b + h * h));
        }
        else if (choice == 3)
        {
            Console.Write("Enter side: ");
            double s = Convert.ToDouble(Console.ReadLine());
            Console.WriteLine("Area: {0}", s * s);
            Console.WriteLine("Perimeter: {0}", 4 * s);
        }
        else
            break;
    }
  }
}
```

```
// ▯ A) Simple Interest
using System;
class SimpleInterest
{
    static void Main()
    {
        Console.Write("Enter Principal: ");
        float p = float.Parse(Console.ReadLine());
        Console.Write("Enter Rate: ");
        float r = float.Parse(Console.ReadLine());
        Console.Write("Enter Time (in years): ");
        float t = float.Parse(Console.ReadLine());
        float si = (p * r * t) / 100;
        Console.WriteLine("Simple Interest = " + si);
    }
}

// ▯ B) Internship Task Management using Method Overloading
using System;
using System.Collections.Generic;
class Task
{
    public string Title { get; set; }
```

```
    public string Description { get; set; }
    public bool IsComplete { get; set; }
}

class TaskManager
{
    static List<Task> tasks = new List<Task>();

    static void AddTask(string title, string description)
    {
        tasks.Add(new Task { Title = title, Description = description,
IsComplete = false });
    }

    static void AddTask(string title)
    {
        tasks.Add(new Task { Title = title, Description = "", IsComplete =
false });
    }

    static void MarkComplete(string title)
    {
        foreach (var task in tasks)
        {
```

```
            if (task.Title == title)
            {
                task.IsComplete = true;
                break;
            }
        }
    }

    static void DisplayTasks()
    {
        foreach (var task in tasks)
        {
            Console.WriteLine("Title: {0}, Description: {1}, Status: {2}",
task.Title, task.Description, task.IsComplete ? "Completed" :
"Pending");
        }
    }

    static void Main()
    {
        AddTask("Documentation", "Write the report");
        AddTask("Meeting");
        MarkComplete("Meeting");
        DisplayTasks();
```

```
    }
}

// ▯ A) Reverse 3 letters
using System;
class ReverseLetters
{
    static void Main()
    {
        Console.Write("Enter three letters: ");
        char a = Convert.ToChar(Console.ReadLine());
        char b = Convert.ToChar(Console.ReadLine());
        char c = Convert.ToChar(Console.ReadLine());
        Console.WriteLine("Reverse Order: {0} {1} {2}", c, b, a);
    }
}

// ▯ B) Small Library System
using System;
using System.Collections.Generic;
class Book
{
    public string Title;
```

```
    public string Author;
    public bool IsIssued = false;
}

class Member
{
    public string Name;
}

class LibrarySystem
{
    static List<Book> books = new List<Book>();

    static void AddBook(string title, string author)
    {
        books.Add(new Book { Title = title, Author = author });
    }

    static void RemoveBook(string title)
    {
        books.RemoveAll(b => b.Title == title);
    }
```

```
static void IssueBook(string title)
{
    foreach (var book in books)
    {
        if (book.Title == title && !book.IsIssued)
        {
            book.IsIssued = true;
            Console.WriteLine("Book issued.");
            return;
        }
    }
    Console.WriteLine("Book not found or already issued.");
}

static void ReturnBook(string title)
{
    foreach (var book in books)
    {
        if (book.Title == title && book.IsIssued)
        {
            book.IsIssued = false;
            Console.WriteLine("Book returned.");
            return;
```

```
            }
        }
        Console.WriteLine("Book not found or not issued.");
    }

    static void DisplayBooks()
    {
        foreach (var book in books)
            Console.WriteLine("Title: {0}, Author: {1}, Issued: {2}",
book.Title, book.Author, book.IsIssued);
    }

    static void Main()
    {
        AddBook("C# Basics", "James");
        AddBook("OOP", "Smith");
        DisplayBooks();
        IssueBook("OOP");
        DisplayBooks();
        ReturnBook("OOP");
    }
}

// □ A) Frequency of Elements in an Array
```

```
using System;
class FrequencyCounter
{
    static void Main()
    {
        int[] arr = { 1, 2, 2, 3, 3, 3 };
        int[] freq = new int[arr.Length];
        for (int i = 0; i < arr.Length; i++) freq[i] = -1;

        for (int i = 0; i < arr.Length; i++)
        {
            int count = 1;
            for (int j = i + 1; j < arr.Length; j++)
            {
                if (arr[i] == arr[j])
                {
                    count++;
                    freq[j] = 0;
                }
            }
            if (freq[i] != 0)
                freq[i] = count;
        }
```

```
        for (int i = 0; i < freq.Length; i++)
        {
            if (freq[i] != 0)
                Console.WriteLine(arr[i] + " appears " + freq[i] + " times");
        }
    }
}

// □ B) Shopping Cart System
using System;
using System.Collections.Generic;
class Product
{
    public string Name;
    public double Price;
}

class Cart
{
    List<Product> items = new List<Product>();

    public void AddProduct(Product p)
```

```
{
    items.Add(p);
}

public void RemoveProduct(string name)
{
    items.RemoveAll(p => p.Name == name);
}

public void DisplayCart()
{
    double total = 0;
    Console.WriteLine("Cart Items:");
    foreach (var item in items)
    {
        Console.WriteLine("{0} - Rs. {1}", item.Name, item.Price);
        total += item.Price;
    }
    Console.WriteLine("Total Price: Rs. " + total);
}

static void Main()
{
```

```
        Cart cart = new Cart();
        cart.AddProduct(new Product { Name = "Pen", Price = 10 });
        cart.AddProduct(new Product { Name = "Notebook", Price = 50
});
        cart.DisplayCart();
        cart.RemoveProduct("Pen");
        cart.DisplayCart();
    }
}

// □ A) Leap Year Check (again)
using System;
class LeapYearRepeat
{
    static void Main()
    {
        Console.Write("Enter a year: ");
        int year = Convert.ToInt32(Console.ReadLine());
        if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
            Console.WriteLine("Leap Year");
        else
            Console.WriteLine("Not a Leap Year");
    }
}
```

```
// ▯ B) TaskManager with Method Overloading
using System;
using System.Collections.Generic;
class TaskItem
{
    public string Title;
    public string Description;
    public string DueDate;
}

class OverloadedTaskManager
{
    List<TaskItem> taskList = new List<TaskItem>();

    public void AddTask(string title, string description)
    {
        taskList.Add(new TaskItem { Title = title, Description =
description });
    }

    public void AddTask(string title, string description, string dueDate)
    {
```

```
        taskList.Add(new TaskItem { Title = title, Description =
description, DueDate = dueDate });
    }

    public void DeleteTask(string title)
    {
        taskList.RemoveAll(t => t.Title == title);
    }

    public void DisplayTasks()
    {
        foreach (var task in taskList)
            Console.WriteLine("Title: {0}, Description: {1}, Due: {2}",
task.Title, task.Description, task.DueDate);
    }

    static void Main()
    {
        OverloadedTaskManager mgr = new OverloadedTaskManager();
        mgr.AddTask("Design", "Create UI");
        mgr.AddTask("Testing", "Unit Tests", "2025-08-15");
        mgr.DisplayTasks();
        mgr.DeleteTask("Design");
        mgr.DisplayTasks();
```

```
    }
}

// ⍰ A) Find Max and Min in an Array
using System;
class MaxMinArray
{
    static void Main()
    {
        int[] arr = { 20, 50, 10, 90, 30 };
        int max = arr[0], min = arr[0];
        foreach (int val in arr)
        {
            if (val > max) max = val;
            if (val < min) min = val;
        }
        Console.WriteLine("Max = " + max);
        Console.WriteLine("Min = " + min);
    }
}

// ⍰ B) OLA Car Rental System
using System;
```

```
using System.Collections.Generic;
class Car
{
    public string Type;
    public bool IsAvailable = true;
}

class Rental
{
    List<Car> cars = new List<Car>();

    public void AddCar(string type)
    {
        cars.Add(new Car { Type = type });
    }

    public void RemoveCar(string type)
    {
        cars.RemoveAll(c => c.Type == type);
    }

    public void RentCar(string type)
    {
```

```
    foreach (var car in cars)
    {
        if (car.Type == type && car.IsAvailable)
        {
            car.IsAvailable = false;
            Console.WriteLine(type + " rented.");
            return;
        }
    }
    Console.WriteLine("No available " + type);
}

public void ReturnCar(string type)
{
    foreach (var car in cars)
    {
        if (car.Type == type && !car.IsAvailable)
        {
            car.IsAvailable = true;
            Console.WriteLine(type + " returned.");
            return;
        }
    }
```

```
        Console.WriteLine("Car not found or already available");
    }

    public void DisplayCars()
    {
        foreach (var car in cars)
            Console.WriteLine("Type: {0}, Available: {1}", car.Type,
car.IsAvailable);
    }

    static void Main()
    {
        Rental r = new Rental();
        r.AddCar("Sedan");
        r.AddCar("SUV");
        r.DisplayCars();
        r.RentCar("SUV");
        r.DisplayCars();
        r.ReturnCar("SUV");
        r.DisplayCars();
    }
}
// ⍰ A) Sum of Digits in a String
using System;
```

```
class SumOfDigits
{
    static void Main()
    {
        Console.Write("Enter a string of digits: ");
        string input = Console.ReadLine();
        int sum = 0;
        foreach (char c in input)
        {
            if (char.IsDigit(c))
                sum += c - '0';
        }
        Console.WriteLine("Sum of digits: " + sum);
    }
}

// B) PaymentProcessor using Delegates
using System;
delegate void PaymentMethod(double amount);

class PaymentProcessor
{
    static void CreditCard(double amount)
```

```
    {
        Console.WriteLine("Paid Rs. {0} using Credit Card", amount);
    }

    static void DebitCard(double amount)
    {
        Console.WriteLine("Paid Rs. {0} using Debit Card", amount);
    }

    static void PayPal(double amount)
    {
        Console.WriteLine("Paid Rs. {0} using PayPal", amount);
    }

    static void Main()
    {
        PaymentMethod payment;
        Console.Write("Enter amount: ");
        double amt = Convert.ToDouble(Console.ReadLine());

        Console.WriteLine("Choose payment method:\n1. Credit
Card\n2. Debit Card\n3. PayPal");
        int choice = Convert.ToInt32(Console.ReadLine());
```

```
        switch (choice)
        {
            case 1: payment = CreditCard; break;
            case 2: payment = DebitCard; break;
            case 3: payment = PayPal; break;
            default: Console.WriteLine("Invalid"); return;
        }
        payment(amt);
    }
}

// □ A) Basic Calculator using switch-case
using System;
class Calculator
{
    static void Main()
    {
        Console.Write("Enter first number: ");
        double a = Convert.ToDouble(Console.ReadLine());
        Console.Write("Enter second number: ");
        double b = Convert.ToDouble(Console.ReadLine());
        Console.Write("Enter operator (+ - * /): ");
        char op = Convert.ToChar(Console.ReadLine());
```

```
        switch (op)
        {
            case '+': Console.WriteLine("Result: " + (a + b)); break;
            case '-': Console.WriteLine("Result: " + (a - b)); break;
            case '*': Console.WriteLine("Result: " + (a * b)); break;
            case '/': Console.WriteLine(b != 0 ? "Result: " + (a / b) : "Cannot
divide by zero"); break;
            default: Console.WriteLine("Invalid operator"); break;
        }
    }
}

// ⍰ B) LINQ Product Sort by Price
using System;
using System.Collections.Generic;
using System.Linq;
class ProductItem
{
    public string Name;
    public double Price;
}

class ProductSort
```

```
{
    static void Main()
    {
        List<ProductItem> products = new List<ProductItem>
        {
            new ProductItem { Name = "Mouse", Price = 250 },
            new ProductItem { Name = "Keyboard", Price = 700 },
            new ProductItem { Name = "Monitor", Price = 5000 }
        };

        var sorted = products.OrderByDescending(p => p.Price);
        foreach (var p in sorted)
            Console.WriteLine("{0}: Rs. {1}", p.Name, p.Price);
    }
}

// ⍰ B) Stock Price Alert using Delegates
using System;
class Stock
{
    private double _price;
    public delegate void PriceChangedHandler(double newPrice);
    public event PriceChangedHandler PriceChanged;
```

```
    public double Price
    {
        get { return _price; }
        set
        {
            if (_price != value)
            {
                _price = value;
                if (PriceChanged != null)
                    PriceChanged(_price);
            }
        }
    }
}

class StockTest
{
    static void Notify(double newPrice)
    {
        Console.WriteLine("Stock price changed to Rs. " + newPrice);
    }
```

```
    static void Main()
    {
        Stock s = new Stock();
        s.PriceChanged += Notify;
        s.Price = 100;
        s.Price = 120;
    }
}
```